

Length Extension Attack

Vulnerable : All Merkle–Damgård-based hash functions

so MD5, SHA-1 and SHA-2 (but not SHA-3)



Flickr's API Signature Forgery Vulnerability

Thai Duong and Julian Rizzo

Date Published: Sep. 28, 2009

Advisory ID: MOCB-01

Advisory URL: http://netifera.com/research/flickr_api_signature_forgery.pdf

Title: Flickr's API Signature Forgery Vulnerability

Remotely Exploitable: Yes

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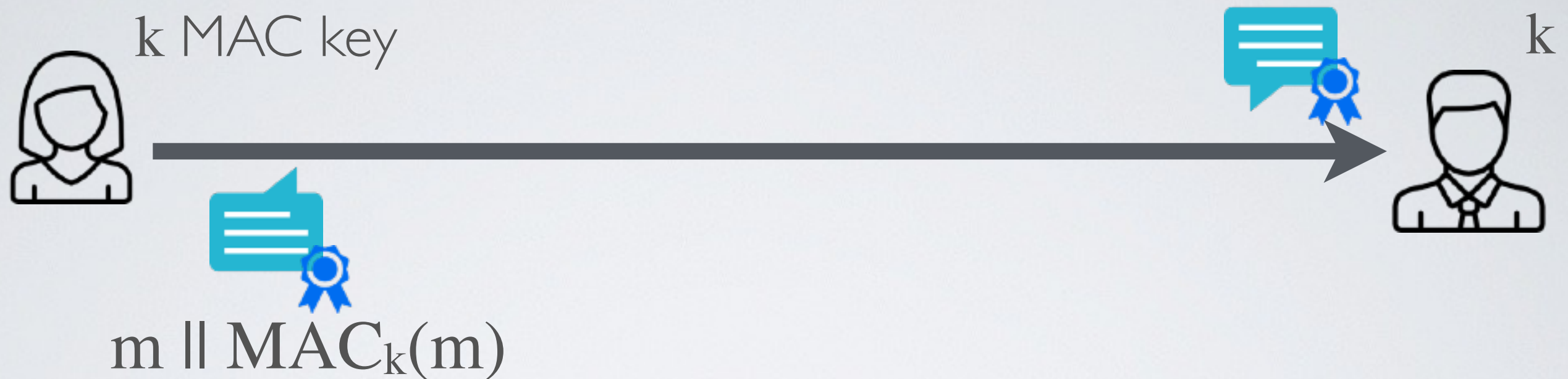
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Good HMAC



Alice and Bob share a key k

➡ Option 1 : envelope method

$$\text{MAC}_k(m) = H(k \parallel m \parallel k)$$

➡ Option 2 : padding method (i.e. HMAC standard)

$$\text{HMAC}_k(m) = H((k \oplus \text{opad}) \parallel H((k \oplus \text{ipad}) \parallel m))$$